

Project Proposal:

Group 10

ESC113M Term Project - 2025 Even Semester

2nd April 2025

Numerical Analysis of a Simplified FCC Reactor Model

1) Authors

Shlok Sharma

240988

shlokns24@iitk.ac.in

Siddhant Tomer

241012

siddhantt24@iitk.ac.in

Shivansh Pandey

240982

shivanshp24@iitk.ac.in

Satvik Mittal

240943

satvikm24@iitk.ac.in

Saloni Kumari

240915

saloni24@iitk.ac.in

Pallav Pratibh

240723

pallavp24@iitk.ac.in

Swastik Bhardwaj

241075

swastikb24@iitk.ac.in

Arman Singh

240183

armank24@iitk.ac.in

Nishant Sharma

240707

nishants24@iitk.ac.in

Akash Lodhi

240079

akashl24@iitk.ac.in

2) Introduction

Fluidized Catalytic Cracking (FCC) is a key process in petroleum refining, breaking down heavy hydrocarbons into valuable lighter products. This project aims to numerically analyze a simplified FCC riser reactor model using mass and energy balances. We will compare the accuracy of *Explicit Euler's method* and *4th-order Runge-Kutta (RK4)* for solving the governing equations.

3) Objectives

- ❖ Develop and solve a basic FCC reactor model numerically.
- ❖ Compare *Euler's method* and *RK4* in terms of accuracy and stability.
- ❖ Analyze the effect of key parameters on reactor performance.

4) Mathematical Model

We consider the following equations:

Mass Balance (Hydrocarbon Conversion): $\frac{dX}{dZ} = k(T)A(1 - X)$

Energy Balance (Temperature Profile): $\frac{dT}{dZ} = \frac{\Delta H_r \cdot k(T)A(1-X)}{\rho C_p}$

Catalyst Deactivation: $\frac{dA}{dZ} = -k_d(T)A$

5) Variable Definitions

X: Fractional hydrocarbon conversion ($0 \leq X \leq 1$)
T: Temperature (K)
A: Catalyst activity (dimensionless, $0 \leq A \leq 1$)
Z: Axial position along the reactor height (m)
k(T): Reaction rate constant (1/s), temperature-dependent via Arrhenius equation:

$$k(T) = k_0 e^{-Ea/RT}$$

k_d(T): Catalyst deactivation rate constant (1/s), also temperature-dependent:

$$k_d(T) = k_{d0} e^{-Ead/RT}$$

ΔH_r: Heat of reaction (J/mol)

- ρ : Gas density (kg/m³)
- C_p : Heat capacity (J/kg·K)
- R : Universal gas constant (J/mol·K)
- E_a : Activation energy for reaction (J/mol)
- E_{ad} : Activation energy for catalyst deactivation (J/mol)
- k_0 : Pre-exponential factor for reaction rate (1/s)
- k_{d0} : Pre-exponential factor for catalyst deactivation (1/s)

6) Numerical Approach

- ❖ Solve coupled ODE using *Explicit Euler's method* and *RK4* on vectorized equations.
- ❖ Compare accuracy of both the methods by plotting curves.

7) Initial Conditions & Parameters

Initial Conditions:

- $X(0) = 0$ (No initial conversion)
- $T(0) = T_{feed}$ (Feed temperature)
- $A(0) = 1$ (Fresh catalyst activity)

Varying Parameters:

- ❖ Reaction rate constant pre-factor k_0
- ❖ Catalyst deactivation pre-factor k_{d0}
- ❖ Activation energies E_a, E_{ad}
- ❖ Heat of reaction ΔH_r
- ❖ Reactor length Z

8) Expected Outcomes

Calculated Values:

- ❖ Conversion $X(Z)$ along the reactor height.
- ❖ Temperature $T(Z)$ profile along the reactor height.
- ❖ Catalyst activity $A(Z)$ degradation along the reactor height.

- ❖ Error estimation for both numerical methods (Euler vs. RK4).

9) **References**

- 1) Weekman, V. E. (1968). 'A Model of Catalytic Cracking Conversion in a Riser Reactor.' AIChE Journal.
- 2) Yui, S. M., & San Miguel, G. (1989). 'Catalyst Deactivation in FCC Units.' Fuel Processing Technology.
- 3) Dorfman, L. A. (2013). 'Numerical Methods with Chemical Engineering Applications.' CRC Press.